系统对接，在线导入方式

1、结构化信息(data.json)

格式为geojson，字段参考如下

代码块

```
{"type":"Feature","geometry":{"coordinates":[[[106.5441769383298,29.562370566536227],[106.54418767002979,29.562371733124348],[106.54419214152978,29.56236881704467],[106.5441954952298,29.562356374478682],[106.54419929602977,29.562349958633476],[106.54419974322978,29.562343348226232],[106.54419325942978,29.562340237584085],[106.5441769383298,29.562370566536227]]],"type":"Polygon"},"properties":{"date":1732614339113,"severity":"DL_MEDIUM","area":3.603916120053327,"typeName":"内部病害","disCode":4,"archiveId":"","disType":"疏松","imageUrl":"","name":"L-T-16","id":197198,"status":"UNCHANGED"}}
{"type":"Feature","geometry":{"coordinates":[[[106.54466691205968,29.5615054552898],[106.54465707465967,29.561495150469348],[106.54465416815967,29.561476486458336],[106.54464030635968,29.561462682620466],[106.5446503673597,29.56144440616747],[106.54466735925968,29.561449461689417],[106.54467429015969,29.561468903958698],[106.54467496085971,29.56148776253439],[106.54466691205968,29.5615054552898]]],"type":"Polygon"},"properties":{"date":1732615041671,"severity":"DL_MEDIUM","area":12.669972119598242,"typeName":"内部病害","disCode":4,"archiveId":"","disType":"疏松","imageUrl":"","name":"L-D-14","id":197199,"status":"UNCHANGED"}}
```

12.25更新，增加disCode字段

data (1).json
156.32KB

枚举字典

disType:

表观病害

内部病害

工程范围

工程范围为最新添加的geojson，添加到最后一行，表示作业的覆盖范围。

严重程度：

代码块

```
  // [Description("严重")]
DL_WORST,
    //    [Description("一般")]
DL_MEDIUM,
      //  [Description("轻微")]
DL_NORMAL,
DL_NULL;
```

病害类型：

类别	名称	标签	代码
机场水泥	CRACKS	横向裂缝、纵向裂缝、斜向裂缝	
	CORNER FRACTURE	角隅断裂	
	CRUSHING PLATES	破碎板、交叉裂缝	
	SUBSIDENCE	沉陷、错台	
	BURST	胀裂	
	JOINT SEALANT DAMAGE	填缝料损坏	
	SEAM BROKEN	接缝破碎	
	PUMPING	唧泥、板底脱空	
	DURABILITY CRACKS	耐久性裂缝	
	CONTRACTIBILITY CRACK	收缩裂缝	
	POT HOLE	坑洞	
	PEELING	起皮、龟裂、细微裂纹	
	CORNER SPALLING	板角剥落	
	SMALL PATCHES	小补丁	

	LARGE PATCHES	大补丁	
	ASR	碱硅酸反应裂缝	
机场沥青	CROCODILE CRACKS	龟裂	
	ZIGZAG CRACKS	不规则裂缝	
	CRACKS	纵向裂缝、横向裂缝	
	REFLECTIVE CRACK	反射裂缝	
	SLIP CRACK	滑移裂缝	
	LOOSENESS AND AGING	松散、老化	
	FLUSHING ASPHALT	泛油	
	AGGREGATE POLISHING	集料磨光	
	SUBSIDENCE	沉陷	
	UPLIFT	隆起	
	WHEEL TRACKING	轮辙	
	WASHBOARD	搓板	
	SQUEEZING	推挤	
	JET ABLATION	喷气烧蚀	
	OIL CORROSION	油料腐蚀	
	PATCHES AND DIGGING PATCHES	补丁、开挖补块	
公路水泥	SHATTEREDSLAB	破碎板	
	CRACKING	裂缝	
	CORNERBREAK	板角断裂	
	FAULTING	错台	
	BLOWUPS	拱起	
	CORNERSPALLING	边角剥落	
	JOINTSEALDAMAGE	接缝料损坏	
	POPOUT	坑洞	
	PUMPING	唧泥	
	SPALLING	露骨	
	PATCH	修补	
公路沥青	ALLIGATORCRACK	龟裂	
	BLOCKCRACK	块状裂缝	
	LONGITUDINALCRACKI	纵向裂缝	

	NG	纵向裂缝	
	TRANSVERSECRACKING	横向裂缝	
	SETTLING	沉陷	
	RUTTING	车辙	
	RIPPINGANDSHOVING	波浪拥包	
	POTHOLES	坑槽	
	RAVELING	松散	
	BLEEDING	泛油	
	PATCH	修补	
军用机场水泥	CRACKS	板块碎裂	
	CORNER FRACTURE	板块断裂	
	CRUSHING PLATES	板角断裂	
	SUBSIDENCE	板块松动	
	BURST	板间错台	
	JOINT SEALANT DAMAGE	道面变形	
	SEAM BROKEN	边角剥落	
	PUMPING	表面剥落	
	DURABILITY CRACKS	表面裂缝	
	CONTRACTIBILITY CRACK	补丁补块	
军用机场沥青	CROCODILE CRACKS	龟裂	
	ZIGZAG CRACKS	不规则裂缝	
	CRACKS	纵向裂缝、横向裂缝	
	REFLECTIVE CRACK	反射裂缝	
	SLIP CRACK	滑移裂缝	
	LOOSENESS AND AGING	松散、老化	
	FLUSHING ASPHALT	泛油	
	AGGREGATE POLISHING	集料磨光	
	SUBSIDENCE	沉陷	
	UPLIFT	隆起	

	WHEEL TRACKING	轮辙	
	WASHBOARD	搓板	
	SQUEEZING	推挤	
	JET ABLATION	喷气烧蚀	
	OIL CORROSION	油料腐蚀	
	PATCHES AND DIGGING PATCHES	补丁、开挖补块	

diseaseType.xml
8.96KB

内部病害

代码块

```
switch (this) {
    case DT_GAP:
        return "脱空";
    case DT_WATER:
        return "富含水";
    case DT_VOID:
        return "空洞";
    case DT_CRACK:
        return "裂缝";
    case DT_LACUNAS:
        return "疏松";
    case DT_DELAMIN:
        return "层离";
    case DT_SUBSIDENCE:
        return "沉陷";
    default:
        return "未知";

}
```

病害状态字典

代码块

```
public enum DiseaseStatus {
    UNCHANGED,
    NEW,
    REPAIRED,
    WORSEN,
    CHANGED,
    REAPPEAR,
    NULL,
}
```

2、图层信息(layer.json)

代码块

```
[
            {
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/das-geoserver/geoserver/gwc/service/wmts?CQL_FILTER=project_id In (103919)&ENV=&layer=penghui:data_db_new_surf&style=&tilematrixset=3857&Service=WMTS&Request=GetTile&Version=1.0.0&Format=image/png&TileMatrix=3857:{z}&TileCol={x}&TileRow={y}",
                "name": "表观病害"
            },
            {
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/das-geoserver/geoserver/gwc/service/wmts?CQL_FILTER=project_id In (103919)&ENV=&layer=penghui:data_db_new_inner&style=&tilematrixset=3857&Service=WMTS&Request=GetTile&Version=1.0.0&Format=image/png&TileMatrix=3857:{z}&TileCol={x}&TileRow={y}",
                "name": "内部病害"
            },
            {
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/image/103919/image/tile/{z}/{x}/{y}.png",
                "name": "实景地图"
            }
        ]
```

实例demo，高德底图偏移和图层加载，leaflet框架,该代码可直接打开加载。

代码块

```
<!DOCTYPE html>
<html>
<head>
    <title>Leaflet 高德地图示例</title>
    <link rel="stylesheet" href="https://unpkg.com/leaflet/dist/leaflet.css" />
    <style>
        #map { height: 100vh; } /* 设置地图容器高度为全屏 */
    </style>
</head>
<body>
    <div id="map"></div> <!-- 地图容器 -->
    <script src="https://unpkg.com/leaflet/dist/leaflet.js"></script>
    <script>
        // 初始化地图，设置起始位置和缩放级别
        var map = L.map('map').setView([30.67776629150997, 104.15970804286191], 13);

        // 添加高德地图作为底图
        var amapLayer = L.tileLayer('https://webrd0{s}.is.autonavi.com/appmaptile?lang=zh_cn&size=1&scale=1&style=8&x={x}&y={y}&z={z}', {
            subdomains: ['1', '2', '3', '4'],
            attribution: '© 高德地图',
                        coordType: 'gcj02',
        }).addTo(map);

        // 图层数据
        var layers = [
            {
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/das-geoserver/geoserver/gwc/service/wmts?CQL_FILTER=project_id In (103919)&ENV=&layer=penghui:data_db_new_surf&style=&tilematrixset=3857&Service=WMTS&Request=GetTile&Version=1.0.0&Format=image/png&TileMatrix=3857:{z}&TileCol={x}&TileRow={y}",
                "name": "表观病害"
            },
            {
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/das-geoserver/geoserver/gwc/service/wmts?CQL_FILTER=project_id In (103919)&ENV=&layer=penghui:data_db_new_inner&style=&tilematrixset=3857&Service=WMTS&Request=GetTile&Version=1.0.0&Format=image/png&TileMatrix=3857:{z}&TileCol={x}&TileRow={y}",
                "name": "内部病害"
            },
            {
```

```
                "type": "WMTS",
                "url": "https://home.gm-robot.com:9104/image/103919/image/tile/{z}/{x}/{y}.png",
                "name": "实景地图"
            }
        ];

        // 遍历图层数据并添加到地图
        var overlayLayers = {};
        layers.forEach(function(layer) {
            if (layer.type === "WMTS") {
                var tileLayer = L.tileLayer(layer.url, {
                    attribution: layer.name,
                    maxZoom: 18,
                    minZoom: 1
                });
                overlayLayers[layer.name] = tileLayer;
            }
        });

        // 添加图层控制
        L.control.layers({ "高德地图": amapLayer }, overlayLayers).addTo(map);


                L.GridLayer.include({

    _setZoomTransform: function (level, center, zoom) {

        if (center != undefined && this.options) {

            // 把鼠标坐标由WGS84坐标系转为DB09或GCJ02坐标系，leaflet从图面上获取的坐标是WGS84坐标系的，需要转换到瓦片对应的坐标系，才能匹配瓦片
            if (this.options.coordType == 'gcj02') {

                center = L.coordTransform().wgs84_To_gcj02(center.lng, center.lat);

            } else if (this.options.coordType == 'bd09') {

                center = L.coordTransform().wgs84_To_bd09(center.lng, center.lat);

            }else if(this.options.coordType == 'dragImg'){
                center = {lng:center.lng-this.options.transformParam.lng, lat:center.lat-this.options.transformParam.lat};
            }

```

```
        }

        let viewHalf = this._map.getSize().divideBy(2);

        let mapProject = this._map.project(center, zoom);

        let mapPanePos = this._map._mapPane._leaflet_pos;

        let newPixelOrigin = mapProject.subtract(viewHalf)._add(mapPanePos).round();

        let scale = this._map.getZoomScale(zoom, level.zoom),

            translate = level.origin.multiplyBy(scale)

            .subtract(newPixelOrigin).round();


        if (L.Browser.any3d) {

            L.DomUtil.setTransform(level.el, translate, scale);

        } else {

            L.DomUtil.setPosition(level.el, translate);

        }

    },

    // 获取地图容器像素四至，根据修改后的鼠标中心点获取

    _getTiledPixelBounds: function (center) {


        if (center != undefined && this.options) {

            if (this.options.coordType == 'gcj02') {

                center = L.coordTransform().wgs84_To_gcj02(center.lng, center.lat);

            } else if (this.options.coordType == 'bd09') {

                center = L.coordTransform().wgs84_To_bd09(center.lng, center.lat);

```

```
            }

        }

        let map = this._map,

            mapZoom = map._animatingZoom ? Math.max(map._animateToZoom, map.getZoom()) : map.getZoom(),

            scale = map.getZoomScale(mapZoom, this._tileZoom),

            pixelCenter = map.project(center, this._tileZoom).floor(),

            halfSize = map.getSize().divideBy(scale * 2);

        return new L.Bounds(pixelCenter.subtract(halfSize), pixelCenter.add(halfSize));

    }

});


// 坐标系转换
L.CoordTransform = function () {

    //bd09转wgs84

    this.bd09_To_wgs84 = function(lng, lat) {

        var gcj02 = this.bd09_To_gcj02(lng, lat);

        var wgs84 = this.gcj02_To_wgs84(gcj02.lng, gcj02.lat);

        return wgs84;

    }

    //wgs84转bd09

    this.wgs84_To_bd09 = function (lng, lat) {

        var gcj02 = this.wgs84_To_gcj02(lng, lat);

        var bd09 = this.gcj02_To_bd09(gcj02.lng, gcj02.lat);

        return bd09;
```

```
    }

    //wgs84转gcj02

    this.wgs84_To_gcj02 = function (lng, lat) {

        var dLat = transformLat(lng - 105.0, lat - 35.0);

        var dLng = transformLng(lng - 105.0, lat - 35.0);

        var radLat = lat / 180.0 * pi;

        var magic = Math.sin(radLat);

        magic = 1 - ee * magic * magic;

        var sqrtMagic = Math.sqrt(magic);

        dLat = (dLat * 180.0) / ((a * (1 - ee)) / (magic * sqrtMagic) * pi);

        dLng = (dLng * 180.0) / (a / sqrtMagic * Math.cos(radLat) * pi);

        var mgLat = lat + dLat;

        var mgLng = lng + dLng;

        var newCoord = {

            lng: mgLng,

            lat: mgLat

        };

        return newCoord;

    }

    //gcj02转wgs84

    this.gcj02_To_wgs84 = function (lng, lat) {

        var coord = transform(lng, lat);

        var lontitude = lng * 2 - coord.lng;

```

```
        var latitude = lat * 2 - coord.lat;

        var newCoord = {

            lng: lontitude,

            lat: latitude

        };

        return newCoord;

    }

    //gcj02转bd09

    this.gcj02_To_bd09 = function (lng, lat) {

        var z = Math.sqrt(lng * lng + lat * lat) + 0.00002 * Math.sin(lat * x_pi);

        var theta = Math.atan2(lat, lng) + 0.000003 * Math.cos(lng * x_pi);

        var bd_lng = z * Math.cos(theta) + 0.0065;

        var bd_lat = z * Math.sin(theta) + 0.006;

        var newCoord = {

            lng: bd_lng,

            lat: bd_lat

        };

        return newCoord;

    }

    //bg09转gcj02

    this.bd09_To_gcj02 = function (lng, lat) {

        var x = lng - 0.0065;

        var y = lat - 0.006;

```

```
        var z = Math.sqrt(x * x + y * y) - 0.00002 * Math.sin(y * x_pi);

        var theta = Math.atan2(y, x) - 0.000003 * Math.cos(x * x_pi);

        var lontitude = z * Math.cos(theta);

        var latitude = z * Math.sin(theta);

        var newCoord = {

            lng: lontitude,

            lat: latitude

        };

        return newCoord;

    }



    var pi = 3.14159265358979323846 26;

    var a = 6378245.0;

    var ee = 0.00669342162296594323;

    var x_pi = pi * 3000.0 / 180.0;

    var R = 6378137;



    function transform(lng, lat) {

        var dLat = transformLat(lng - 105.0, lat - 35.0);

        var dLng = transformLng(lng - 105.0, lat - 35.0);

        var radLat = lat / 180.0 * pi;

        var magic = Math.sin(radLat);

        magic = 1 - ee * magic * magic;

        var sqrtMagic = Math.sqrt(magic);
```

```

        dLat = (dLat * 180.0) / ((a * (1 - ee)) / (magic * sqrtMagic) * pi);

        dLng = (dLng * 180.0) / (a / sqrtMagic * Math.cos(radLat) * pi);

        var mgLat = lat + dLat;

        var mgLng = lng + dLng;

        var newCoord = {

            lng: mgLng,

            lat: mgLat

        };

        return newCoord;

    }

    function transformLat(x, y) {

        var ret = -100.0 + 2.0 * x + 3.0 * y + 0.2 * y * y + 0.1 * x * y + 0.2 * Math.sqrt(Math.abs(x));

        ret += (20.0 * Math.sin(6.0 * x * pi) + 20.0 * Math.sin(2.0 * x * pi)) * 2.0 / 3.0;

        ret += (20.0 * Math.sin(y * pi) + 40.0 * Math.sin(y / 3.0 * pi)) * 2.0 / 3.0;

        ret += (160.0 * Math.sin(y / 12.0 * pi) + 320 * Math.sin(y * pi / 30.0)) * 2.0 / 3.0;

        return ret;

    }



    function transformLng(x, y) {

        var ret = 300.0 + x + 2.0 * y + 0.1 * x * x + 0.1 * x * y + 0.1 * Math.sqrt(Math.abs(x));

```

```
        ret += (20.0 * Math.sin(6.0 * x * pi) + 20.0 * Math.sin(2.0 * x * pi)) * 2.0 / 3.0;

        ret += (20.0 * Math.sin(x * pi) + 40.0 * Math.sin(x / 3.0 * pi)) * 2.0 / 3.0;

        ret += (150.0 * Math.sin(x / 12.0 * pi) + 300.0 * Math.sin(x / 30.0 * pi)) * 2.0 / 3.0;

        return ret;

    }

}



L.coordTransform = function () {

    return new L.CoordTransform();

}
    </script>
</body>
</html>
```

例子，迁移到高德sdk的瓦片加载方式

test2.html
8.70KB